

Incompressible fluids III: Steady Navier-Stokes

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From Stokes to Navier–Stokes

We now return to the incompressible fluid equations.

Compared with Stokes, the new difficulty is the nonlinear convective term

$$(u \cdot \nabla)u.$$

The relevant nondimensional parameter is the Reynolds number

$$Re = \frac{UL}{\nu},$$

where U and L are characteristic velocity and length scales.

- ▶ $Re \ll 1$: diffusion dominates, flow is close to Stokes,
- ▶ $Re \gtrsim 1$: convection becomes important,
- ▶ large Re : transition to turbulence and strong multiscale effects.

Remark. This is why CFD for Navier–Stokes is much more delicate than for Stokes.

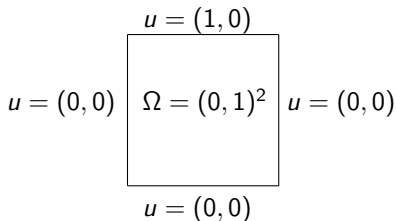
Steady incompressible Navier–Stokes system

Assume constant viscosity $\nu > 0$ and steady state. The strong form is

$$\begin{cases} -\nu \Delta u + (u \cdot \nabla)u + \nabla p = f & \text{in } \Omega, \\ \nabla \cdot u = 0 & \text{in } \Omega, \\ u = g_D & \text{on } \partial\Omega. \end{cases}$$

with $(u \cdot \nabla)u = \sum_{i=1}^d u_i D_i u = [u \cdot \nabla u_j]_j$.

A standard benchmark: the **lid-driven cavity flow**:



A square domain $\Omega = (0, 1)^2$ is shown. The velocity boundary conditions are:

- Top boundary: $u = (1, 0)$
- Bottom boundary: $u = (0, 0)$
- Left boundary: $u = (0, 0)$
- Right boundary: $u = (0, 0)$

Classical test case for comparing discretizations and nonlinear solvers.

Weak formulation

For homogeneous Dirichlet boundary conditions, set

$$V := [H_0^1(\Omega)]^d, \quad Q := L_0^2(\Omega) := \left\{ q \in L^2(\Omega) : \int_{\Omega} q = 0 \right\}.$$

Then the weak problem is: find $(u, p) \in V \times Q$ such that

$$\begin{cases} a(u, v) + c(u; u, v) + b(v, p) = (f, v) & \forall v \in V, \\ b(u, q) = 0 & \forall q \in Q, \end{cases}$$

where

$$a(u, v) = \nu \int_{\Omega} \nabla u : \nabla v \, dx, \quad b(v, q) = - \int_{\Omega} q \nabla \cdot v \, dx,$$

$$c(w; u, v) = \int_{\Omega} (w \cdot \nabla u) \cdot v \, dx.$$

The pressure again acts as a Lagrange multiplier enforcing incompressibility.

Continuity of the convection term

The trilinear form is

$$c(w; u, v) = \sum_{i,j=1}^d \int_{\Omega} w_j \partial_j u_i v_i dx.$$

Using Hölder and the Sobolev embedding $H^1(\Omega) \hookrightarrow L^4(\Omega)$ for $d = 2, 3$,

$$|c(w; u, v)| \leq \|w\|_{L^4(\Omega)} \|\nabla u\|_{L^2(\Omega)} \|v\|_{L^4(\Omega)} \leq \hat{C} \|w\|_{H^1(\Omega)} \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}.$$

So c is continuous on $[H^1(\Omega)]^d$ for $d = 2, 3$.

Moreover, if $\nabla \cdot w = 0$ and $w \cdot n = 0$ on $\partial\Omega$, then

$$c(w; v, v) = 0.$$

This skew-symmetry is the key energy property of the convective term.

Well-posedness remarks

Unlike Stokes, the steady Navier–Stokes problem is nonlinear, so the theory is more delicate.

- ▶ Existence can be proved under standard assumptions in 2D, and under suitable hypotheses in 3D.
- ▶ Uniqueness typically requires a small-data condition, for example forcing small relative to viscosity (see later).
- ▶ The irrotational part of the force can be absorbed into the pressure, so one may restrict attention to the divergence-free part of f .

The analytical difficulties come from the same convective term. In turns, this also creates numerical difficulties at moderate/high Reynolds number.

Operator form and relation with Stokes

Introduce the nonlinear operator $A(w) : V \rightarrow V'$ by

$$\langle A(w)u, v \rangle := a(u, v) + c(w; u, v), \quad \forall u, v \in V,$$

and the divergence operator $B : V \rightarrow Q'$ by

$$\langle Bv, q \rangle := b(v, q).$$

Let $\ell \in V'$ be defined by

$$\langle \ell, v \rangle = (f, v).$$

Then Navier–Stokes can be written as

$$\begin{cases} A(u)u + B^t p = \ell & \text{in } V', \\ Bu = 0 & \text{in } Q'. \end{cases}$$

This has the same mixed structure as Stokes, but with a nonlinear velocity operator.

Reduction to the divergence-free space

As in the linear case, let

$$Z := \ker B = \{v \in V : \nabla \cdot v = 0\}.$$

If the inf-sup condition holds, then one may think of solving first for the velocity in the divergence-free space:

$$\text{Find } u \in Z \text{ such that } \tilde{a}(u; u, v) = \langle \ell, v \rangle \quad \forall v \in Z, \quad (1)$$

where

$$\tilde{a}(w; u, v) := a(u, v) + c(w; u, v).$$

Once u is known, the pressure can be recovered as in the Stokes case.

Theorem

Let f be such that $\nabla \cdot f = 0$ and $\|f\|_{L^2(\Omega)} < \frac{\nu^2}{\hat{C} C_\Omega^{1/2}}$, with C_Ω the Poincaré inequality constant. Then, problem (1) has a unique solution in Z .

Practical remark. Exactly divergence-free discrete spaces are difficult to construct, so in practice one usually discretizes the full mixed system.

Discretization

We use again the the Taylor–Hood element. For $k \geq 1$,

$$V_h = \{v_h \in [H_0^1(\Omega)]^d : v_h|_K \in [\mathbb{P}_{k+1}(K)]^d\} \subset [H_0^1(\Omega)]^d,$$

$$Q_h = \{q_h \in H_0^1(\Omega) : q_h|_K \in \mathbb{P}_k(K)\} \subset L_0^2(\Omega).$$

Taylor-Hood FEM: find $(u_h, p_h) \in V_h \times Q_h$ such that

$$a(u_h, v_h) + c(u_h; u_h, v_h) + b(v_h, p_h) = (f, v_h) \quad \forall v_h \in V_h,$$

$$b(u_h, q_h) = 0 \quad \forall q_h \in Q_h.$$

The main **additional issue**, compared with Stokes, is no longer only the inf-sup condition, but also the **nonlinear solve**.

Linearize and then solve: Newton's method

Define the nonlinear residual map

$$F(u, p) = \begin{pmatrix} -\nu \Delta u + (u \cdot \nabla) u + \nabla p - f \\ -\nabla \cdot u \end{pmatrix}.$$

We seek a zero of F .

Given an initial guess $X^0 = (u^0, p^0)$, Newton's method computes $X^{k+1} = X^k + \delta X^k$ by solving

$$DF(X^k) \delta X^k = -F(X^k).$$

That is, one solves for the increment

$$\delta X^k = (\delta u^k, \delta p^k).$$

So the key step is the computation of the Jacobian of F .

Jacobian and linearized Navier–Stokes system

To solve $DF(X^k) \delta X^k = -F(X^k)$, we need to evaluate the Jacobian of F along δX^k at x^k , that is the Gâteaux derivative at $x^k = (u^k, p^k)$:

Jacobian and linearized Navier–Stokes system

The Gâteaux derivative at (u^k, p^k) is

$$DF(u^k, p^k)(\delta u^k, \delta p^k) = \begin{pmatrix} -\nu \Delta \delta u^k + (u^k \cdot \nabla) \delta u^k + (\delta u^k \cdot \nabla) u^k + \nabla \delta p^k \\ -\nabla \cdot \delta u^k \end{pmatrix}.$$

Hence the Newton step solves

$$-\nu \Delta \delta u^k + (u^k \cdot \nabla) \delta u^k + (\delta u^k \cdot \nabla) u^k + \nabla \delta p^k = -F_1(u^k, p^k),$$

$$-\nabla \cdot \delta u^k = \nabla \cdot u^k, \quad \delta u^k = 0 \text{ on } \partial\Omega$$

(for fixed Dirichlet data).

This is a linear **Oseen**-type saddle-point problem at each Newton step.

Initial guess and continuation

Initial guess is essential¹:

Continuation method (standard strategy):

1. solve the corresponding Stokes problem,
2. use this as initial guess for a Navier–Stokes solve at moderate viscosity,
3. progressively decrease the viscosity (or increase the Reynolds number), always using the previous solution as initial guess.

This is particularly effective for benchmark problems such as the lid-driven cavity.

¹Newton converges rapidly only near the solution.

Circumventing inf-sup: stabilized FEMs

Unstable Stokes pairs such as

- ▶ P_1-P_0 (discontinuous pressures),
- ▶ equal-order continuous pairs such as P_1-P_1 ,

do not satisfy a mesh-uniform inf-sup condition.

A standard remedy is to add a stabilization term. For Stokes, a basic Galerkin least-squares type stabilization modifies the bilinear form and right-hand side:

$$B_h((u, p), (v, q)) := B((u, p), (v, q)) + \delta \sum_{K \in \mathcal{T}_h} (-\Delta u + \nabla p, -\Delta v + \nabla q)_K,$$

$$F_h(v, q) := (f, v) + \delta \sum_{K \in \mathcal{T}_h} (f, -\Delta v + \nabla q)_K.$$

Here δ is a stabilization parameter depending on h .

Why stabilization restores coercivity

In matrix form, the stabilized problem has the same saddle-point structure but with a nonzero $(2, 2)$ block:

$$\mathbb{B}_h = \begin{pmatrix} A & B^T \\ -B & C \end{pmatrix}.$$

For equal-order elements, taking $(v, q) = (u, p)$ gives

$$\mathbb{B}_h((u, p), (u, p)) = a(u, u) + \delta(\nabla p, \nabla p)_\Omega,$$

so that

$$\mathbb{B}_h((u, p), (u, p)) = \|\nabla u\|_{L^2(\Omega)}^2 + \delta \|\nabla p\|_{L^2(\Omega)}^2.$$

Hence the stabilized bilinear form becomes coercive, and the discrete problem can be treated by Lax–Milgram.

Remark. The coercivity constant depends on δ , hence on h ; the parameter must be chosen large enough for stability but small enough not to destroy the convergence rate.

Why this matters for Navier–Stokes

The same applies for Navier–Stokes:

- ▶ the convective term may require stabilization, as in advection-dominated transport,
- ▶ the velocity–pressure pair must either satisfy inf–sup or be stabilized,
- ▶ practical CFD also depends on time stepping, linearization, and initialization.

At low Reynolds number one may work with inf–sup stable pairs and Newton or Picard iteration. At moderate and large Reynolds number, stabilization and turbulence modelling become central ingredients.